

ASUTOSH PANDA

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SKILLS

GPU & AI Infrastructure: NVIDIA GPU Operator, DCGM, drivers/CUDA, MIG, vLLM inference serving, GPU SLOs and observability
Infrastructure & Networking: Kubernetes (GKE, bare-metal), Helm, Kustomize, FluxCD, ArgoCD, Terraform, OpenTofu, Ansible, Linkerd, KubeOne, OpenEBS, GCP, AWS, Hetzner, OVH, Cloudflare
Secrets and Auth: GCP Workload Identity Federation, External Secrets Operator (ESO), GCP Secret Manager, cert-manager, Pomerium
CI/CD and Observability: GitHub Actions, self-hosted ARC runners, Trivy, SonarQube, TruffleHog, Renovate; Prometheus, Grafana, Mimir, Loki, Tempo, Grafana Alloy, grafana-operator, Alertmanager
Development: Golang, Python, FastAPI, Bash, Node.js, TypeScript, PostgreSQL, Linux

PROFESSIONAL EXPERIENCE

ONE2N

Pune, Maharashtra, India

Senior SRE & Team Lead • Client: Fravity AI (Remote)

May 2026 - Present

- Built **fravity-spine**, the production multi-tenant GKE platform powering an AI product: private control plane, CMEK etcd, Workload Identity Federation, OpenTofu across **6** isolated state layers. Replaced bastions and shared SSH with identity-based **kubectrl** via GKE Connect Gateway + Workspace SSO; FluxCD-bootstrapped cert-manager, ESO, Traefik, and Pomerium SSO.
- Led a full-estate infrastructure assessment spanning VMs and **5** GCP plus **5** AWS accounts, cataloguing security, IAM, and cost risks into a prioritised remediation roadmap: surfaced **19** service-account keys, an unowned **32-vCPU** instance, and ~\$**26,520/yr** of zombie spend. Drove the follow-up hardening across the **2**-engineer team I lead: IaC bootstrapped to **60%** across **2** envs, **10** workload SAs and all human IAM migrated to Terraform, CI rebuilt on Workload Identity Federation.
- Delivered the central LGTM observability platform: Mimir + Loki + Tempo (GCS-backed via Workload Identity, **365-day** metrics/logs + **30-day** traces, multi-tenant via X-Scope-OrgID) as authenticated public ingest. Migrated to grafana/alloy single-agent (logs, metrics, traces) across spine and the fravity-dev1 spoke; Alertmanager + Slack via ESO; grafana-operator CRDs for dashboards-as-code.
- Stood up a **5**-tool SAST stack on self-hosted ARC runners (Trivy operator, SonarQube CE, renovate-operator, TruffleHog) to meet the "scans inside our private GKE" audit bar. Shipped **scan-aggregator**, a FastAPI webhook service normalising **7** scanner outputs to a per-repo audit xlsx; first customer-facing deliverable: a customer security review with **76** H/C findings across **4** images.
- Raised platform-team maturity: mentored developers, standardised team tooling (Outline docs, Vaultwarden secrets), and delivered weekly platform status to leadership.
- Delivered staging on **fravity-dev1** GKE as a commit-to-URL GitOps pipeline: CI build, Flux image-automation HelmRelease bumps, a live TLS endpoint, and Slack deploy alerts, all **zero static credentials** via Workload Identity with per-app least-privilege GSAs. Killed staging schema drift by locking the database to **automation-only** writes with **Atlas** declarative migrations as the single source of truth. Root-caused the rollout's stacked failures (HPA runaway, cgroup OOMs, DB networking) and upstreamed the chart fix.

LUXOR TECHNOLOGY

Seattle, WA, US

DevOps Engineer (Remote)

June 2023 - April 2026

- Operated a multi-cluster Kubernetes platform for **20+** teams and **3,000+** enterprise customers: **8+** clusters across Hetzner, OVH, and GCP with **24/7** uptime for L1/L2 validator and RPC workloads (BTC, LTC, Zcash, Stacks, tBTC), including **2** Hetzner bare-metal HA clusters (KubeOne, OpenEBS) provisioned and driver-managed from scratch. Standardised on a **7**-layer FluxCD GitOps pipeline with Linkerd mTLS; migrated **70+** SOPS secrets to GCP Workload Identity.
- Owned the on-call and incident platform: migrated **5** teams from Opsgenie to BetterStack (runbooks, escalation, multi-channel alerting, Prometheus/Grafana-fed). Led response through real outages including a Litecoin hard fork (**zero-downtime** deploy) and a kubelet TLS expiry, and shipped x509 monitoring across **10** clusters with **7-day/24-hour** alerts.
- Owned CI/CD for **50+** repositories on self-hosted ARC runners. Ported a failing Playwright E2E suite from Mac hosted runners (\$**5,000/mo**) to Linux ARC, cutting runtime **40 to 15 min** and cost to \$**600/mo**. Built code-signed Windows + Mac release pipelines and automated FluxCD deployments to GCP Artifact Registry.
- Built org-wide platform tooling: **KubeOrbit** (Go CLI for kubeconfig provisioning across **20+** clusters) and **Memphis** (Python SOC1/2/SOX compliance binary with auditor-ready Drive reports + Linear escalation; solo, **4 weeks** for the planned IPO). Prevented a supply-chain compromise by vendoring Trivy into a composite action across **9** repos; when CVE-2026-33634 later hit aquasecurity/trivy-action, Luxor pipelines were untouched.
- Proposed and built **BVT**, the org's first zero-hardcoded-secrets cluster: replaced SOPS-in-git with ESO + GCP Secret Manager + Workload Identity Federation; the pattern became the gold standard for new clusters. Authored custom Python tooling against Hetzner + OVH firewall APIs to enable secure cross-cloud peering for bare-metal nodes where no Terraform modules existed.

TATA CONSULTANCY SERVICES

Hyderabad, Telangana, India

Systems Engineer (Remote)

October 2020 - June 2023

- Migrated **2,500+** repositories from GitLab, Bitbucket, and SVN to GitHub Enterprise for **120** teams, saving **15 weeks** of manual effort.
- Built PowerShell, Python, and Golang automation across **22** RHEL/WebLogic servers (health checks and storage allocation), saving **380+** man-hours/month.

PROJECTS

- Frit** (May 2026 - Present) 📄 - GPU SRE lab: vLLM on NVIDIA GPU Operator; OSS PRs to **dcgm-exporter** + **gpu-operator**.
- agent-box** (One2N, 2026) - Hardened devcontainer sandbox for AI agents: cap-drop ALL, non-root, no host credentials.
- Platform Zero** (Apr 2026 - Present) 📄 - **4** SRE labs (k8slab, ansiblelab, awslab, gcplab); **0-to-100** in one command.
- Truss** + **Dint** (2026) - Self-hostable BaaS (auth, storage, realtime) and an embeddable user-feedback widget.

TEACHING EXPERIENCE

- Full-Stack Instructor, **4** cohorts (2022-2026), taught NextJS, FastAPI, and DevOps to BSc CS students.

EDUCATION

SASTRA DEEMED UNIVERSITY

Master of Computer Applications (MCA) - Chennai, India - 2021 - 2023